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Question 1 | Correct | Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  // Function to swap two elements
4  void swap(int *a, int *b) {
5      int temp = *a;
6      *a = *b;
7      *b = temp;
8  }
9
10 // Partition function using the last element as the pivot
11 int partition(int arr[], int low, int high) {
12     int pivot = arr[high]; // Pivot element
13     int i = (low - 1);      // Index of smaller element
14
15     for (int j = low; j <= high - 1; j++) {
16         // If current element is smaller than or equal to pivot
17         if (arr[j] <= pivot) {
18             i++; // Increment index of smaller element
19             swap(&arr[i], &arr[j]);
20         }
21     }
22     swap(&arr[i + 1], &arr[high]);
23     return (i + 1);
24 }
25
26 // Quick Sort function (Divide and Conquer)
27 void quickSort(int arr[], int low, int high) {
28     if (low < high) {
29         // pi is partitioning index, arr[p] is now at right place
30         int pi = partition(arr, low, high);
31
32         // Separately sort elements before partition and after partition
33         quickSort(arr, low, pi - 1);
34         quickSort(arr, pi + 1, high);
35     }
36 }
37
38 int main() {
39     int n;
40     scanf("%d", &n);
41
42     int arr[n];
43     for (int i = 0; i < n; i++) {
44         scanf("%d", &arr[i]);
45     }
46
47     // Perform Quick Sort
48     quickSort(arr, 0, n - 1);
49
50     // Print the sorted elements separated by spaces
51     for (int i = 0; i < n; i++) {
52         printf("%d", arr[i]);

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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